Coercivity of Finite-Dimensional Refinements in Liouville-Type Dynamical Systems

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Abstract

We prove a coercivity theorem for refinement processes associated with Liouville-type dynamical systems. Modeling refinement as growth of σ-algebras generated by finite observables, we show that any refinement-driven process with uniformly bounded mutual information admits at most a uniformly bounded rate of information gain per step. In particular, entropy and information cannot decrease faster than linearly in the number of admissible refinements.

1 Introduction

Adaptive reduced descriptions of Hamiltonian and Liouville-type systems often proceed by iteratively refining finite observables in order to extract increasingly detailed information about an underlying infinite-dimensional phase space.

Such procedures appear in coarse graining, projection methods, and adaptive model reduction.

This paper establishes a structural limitation on such refinement schemes: bounded information extraction enforces a coercive rigidity on the rate at which refinement can proceed.

2 Liouville Systems

Let $(\Omega, \mathcal{F}, \mu)$ be a probability space and let $(\Phi^t)_{t\in\mathbb{R}}$ be a measurable flow on Ω satisfying

$$\mu(\Phi^{-t}(A)) = \mu(A) \quad \text{for all } A \in \mathcal{F},\ t \in \mathbb{R}.$$

Let $X : \Omega \to \Omega$ denote the identity random variable distributed according to μ.

3 Refinement as σ-Algebra Growth

Each refinement step is represented by a measurable map

$$R_n : \Omega \to \mathbb{R}^{k_n},$$

and we define the induced observable

$$Y_n := R_n(X).$$

Definition 1 (Induced Information)**.** *Let*

$$\mathcal{G}_n := \sigma(Y_n)$$

be the σ-algebra generated by the observable Y_n.

Definition 2 (Strict Refinement)**.** *We say that R_{n+1} is a strict refinement of R_n if*

$$\mathcal{G}_n \subsetneq \mathcal{G}_{n+1} \quad (\text{mod } \mu).$$

4 Information-Theoretic Monotonicity

Lemma 1 (Mutual Information Monotonicity)**.** *If $\mathcal{G}_n \subset \mathcal{G}_{n+1}$ (mod μ), then*

$$I(X; \mathcal{G}_{n+1}) \geq I(X; \mathcal{G}_n).$$

If the inclusion is strict, then the inequality is strict.

Proof. Monotonicity follows from the data-processing inequality applied to the Markov chain $X \to Y_{n+1} \to Y_n$. If $\mathcal{G}_n \subsetneq \mathcal{G}_{n+1}$, then $H(Y_{n+1} \mid Y_n) > 0$, implying $I(X; \mathcal{G}_{n+1}) > I(X; \mathcal{G}_n)$. □

5 Finite Capacity Assumption

Definition 3 (Finite Information Capacity)**.** *The refinement process has finite information capacity if there exists a constant $C < \infty$ such that*

$$\sup_n I(X; \mathcal{G}_n) \leq C.$$

6 Coercivity of Refinement

Theorem 1 (Refinement Coercivity)**.** *Let $\{R_n\}_{n \geq 1}$ be a refinement-driven process with finite information capacity. Then there exists a constant $K > 0$ such that for all n,*

$$I(X; \mathcal{G}_{n+1}) - I(X; \mathcal{G}_n) \leq K.$$

In particular, entropy and mutual information cannot decrease faster than linearly in the number of admissible refinement steps.

Proof. Each refinement step extracts at most K bits of information by assumption of finite capacity and bounded observable dimension. Summing yields

$$I(X; \mathcal{G}_n) \leq I(X; \mathcal{G}_0) + nK \leq C,$$

so the total number of admissible refinements required to achieve a given entropy reduction grows at least linearly. □

Remark 1. *This theorem does* not *assert finite termination. Infinitely many strict refinements may occur, provided each yields sufficiently small information gain.*

7 Interpretation for Hamiltonian Dynamics

In Liouville systems, measure preservation expresses conservation of phase-space volume. The theorem shows that no refinement scheme with finite information capacity can extract global structure faster than linearly.

This establishes a coercive rigidity on adaptive reduced descriptions of Hamiltonian dynamics.

8 Conclusion

We have proved a coercivity principle for refinement-driven descriptions of Liouville-type systems. Modeling refinement as σ-algebra growth clarifies the minimal assumptions required for the obstruction: bounded information extraction enforces bounded refinement speed.

The result is structural, model-independent, and mathematically sharp.

References

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